

SOFTWARE-DEFINED RADIO With Android Smartphones

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Software-defined radio (SDR) is one of most important wireless communication technologies. It is a unique type of radio system that can tune to any frequency band. The SDR platform is generic, that is, one platform supports various signals of different frequencies.

Earlier, SDR hardware was designed only to support high-end

computer systems and desktop computers. However, with recent advancements in smartphone technology and availability of affordable smartphones, developers have started working on SDR support for mobile phones.

RTL-SDR dongles can easily perform the function of an SDR receiver using an Android smartphone with on-the-go (OTG) support. Fig. 1 shows an RTL-SDR dongle connected to an Android smartphone.

RTL-SDR dongles provide the functionality to receive a signal in the 25MHz-1.75GHz frequency range. The system/radio receiver designed using an RTL-SDR dongle can also be used to study digital communications by electronics and communications engineering students.

An RTL-SDR dongle is mainly

a DVB-T/DVB-T2 signal receiver dongle to view terrestrial TV channels. But developers realised that the RTL2832U chip present in these dongles can also be used as SDR hardware. The dongle is set in test mode and acts as an SDR receiver. DVB-T and DVB-T2 dongles are shown in Fig. 2.

The antenna provided with the dongle (see Fig. 3) is suitable for signal reception. It comes with a magnetic base, so it can be mounted on a metallic surface.

You can use any of the two USB dongles (shown in Fig. 2) for SDR applications. The setup for an Android smartphone based SDR system with only signal reception capability is explained in the following sections.

The SDR system and Android smartphone can be used to perform



Fig. 1: RTL-SDR and Android setup



Fig. 2: RTL-SDR dongle type—DVB-T2 (left) dongle and DVB-T (right) dongle



Fig. 3: Antenna provided with dongle



Fig. 4: Required apps

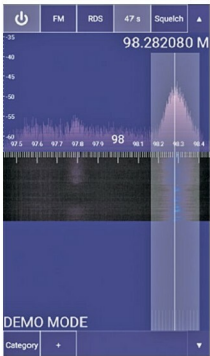


Fig. 5: Interface of SDRTouch app